BUS23-4

Macroeconomics

Measurement, growth, money, and the policy levers that move a whole economy

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Monetary and fiscal policy propagate through a national economy — powerfully, but within limits — and every business decision is taken inside that constraint.

Albert School — 22 hours

Contents

Preface

In **BUS13-4 Microeconomics** you learned to reason about a single agent or a single market: a consumer at a tangency, a firm setting MC = MR, a market clearing where supply meets demand. That reasoning never disappears — it is the foundation everything here rests on — but it answers the wrong question for a manager facing a recession, a rate hike, or a currency swing. Those forces do not live inside one market; they are properties of the **whole system** of markets at once. Macroeconomics is the study of that system: of aggregates — total output, the price level, the unemployment rate, the interest rate — and of the two great levers, **monetary** and **fiscal** policy, that governments and central banks pull to move them.

The course has a single practical promise. By the end you should be able to read a central-bank press release or a national budget **as a business signal**: a rate decision is a change in your cost of capital (the WACC of **BUS33-3**) and in the financing cost of working capital (the BFR of **BUS23-3**); a fiscal stimulus is a shift in the demand your customers bring; an inflation print is a tax on your cash and a clue about what the central bank will do next. The macroeconomy is the constraint layer inside which all the finance and strategy courses operate.

We build the subject in the order understanding actually forms. First we **measure** — GDP, unemployment, inflation — because you cannot reason about what you cannot count, and because every measure hides something (a lesson micro's GDP bridge already flagged). Then we ask what makes an economy **grow** over decades. Then we turn to **money** and the **central bank** that governs it, to **fiscal policy** and its multiplier, and we assemble both into the **IS–LM** framework — the keystone that lets us read any short-run policy episode on a single diagram. We close by putting the framework to work on the three defining shocks of our era: 2008, 2020, and 2022. Throughout, a concrete number comes before every abstraction, and an honest

note on each model's limits comes right after it — because knowing where a model breaks is what separates using it from being used by it.

1 Measuring economic activity

Before any theory, a yardstick. The headline number for an economy's output is **gross domestic product**, which **BUS13-4** defined as the market value of all final goods and services produced within a country in a period. We now make it operational: how it is actually computed, why three different computations must agree, and how to strip out the illusion that rising prices create.

1.1 The three approaches to GDP

> **Worked example — a one-good economy.** Imagine a country that produces only bread. A farmer grows €30 of wheat (using no inputs) and sells it to a baker; the baker turns it into bread sold to households for €100. What is GDP? Not €130 — that double-counts the wheat, which is an **intermediate** input embodied in the bread. GDP is €100, the value of the **final** good. Equivalently, sum the **value added**: €30 by the farmer plus €70 by the baker = €100. And the €100 of bread is bought by households (expenditure €100) and is paid out as €30 + €70 of wages and profit (income €100). All three routes give €100.

Definition 1 GDP can be measured three equivalent ways; they must agree because every euro of output is simultaneously something produced, something bought, and someone's income.

- **Production (value added)**: sum, over all producers, of the value of output minus the value of intermediate inputs.
- **Expenditure**:

$$\text{GDP} = C + I + G + (X - M),$$

  consumption C, investment I (firms' capital spending plus changes in inventories — **not** the purchase of financial assets), government spending on goods and services G, plus net exports, exports X minus imports M.
- **Income**: sum of incomes generated in production — compensation of employees, gross operating surplus (profits), plus taxes on production net of subsidies.

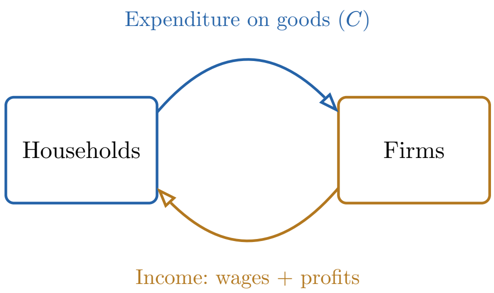

Figure 1: The circular flow behind the identity. What households spend on firms' output (the expenditure route) is what firms produce (the production route) and is paid back out as wages and profits (the income route). The same flow, measured at three points, is why the three approaches give one number.

Common pitfall Two recurring slips. (1) **Intermediate goods**: only final output counts, so the wheat inside the bread is not added on top — count final goods, or value added, never both. (2) "Investment" in the GDP identity means **physical** capital formation (machines, buildings, inventories), not buying shares or bonds. A household buying stock is **saving**, not part of I in $C + I + G + (X - M)$.

1.2 Real versus nominal GDP

Worked example — growth that is only prices. An economy makes 100 loaves at €2 in 2023 (nominal GDP €200) and 100 loaves at €3 in 2024 (nominal GDP €300). Nominal GDP rose 50%, yet not one extra loaf was produced — the entire "growth" is inflation. To see real change we must hold prices fixed: valuing both years at the 2023 price of €2 gives €200 in both years, so **real** GDP growth is zero. That is the number that tracks well-being.

Definition 2 **Nominal GDP** values output at **current** prices; **real GDP** values it at the prices of a fixed **base year**, so changes reflect quantities alone. Their ratio is the price index that deflates one into the other:

$$\text{GDP deflator} = \frac{\text{nominal GDP}}{\text{real GDP}} \times 100.$$

The growth of the deflator is one measure of economy-wide inflation.

Remark Eurostat reports euro-area real GDP in **chain-linked volumes**, which update the base each year and link the results, avoiding the distortion a far-off fixed base introduces. The principle is unchanged: separate quantity from price.

1.3 Labour-market rates

Output is produced by people, and how fully an economy uses its people is measured by three rates that are easy to confuse.

Definition 3 Partition the working-age population. The **labour force** is the employed plus the **unemployed** — those without a job who are available and **actively searching** (the rest are **inactive**: students, retirees, discouraged non-searchers). Then:

$$\text{unemployment rate} = \frac{\text{unemployed}}{\text{labour force}},$$

$$\text{activity (participation) rate} = \frac{\text{labour force}}{\text{working-age population}},$$

$$\text{employment rate} = \frac{\text{employed}}{\text{working-age population}}.$$

Worked example — a town of 1000. Of 1000 working-age people, 600 work, 60 search, 340 are inactive. The labour force is $600 + 60 = 660$. Unemployment rate $= 60/660 \approx 9.1\%$; activity rate $= 660/1000 = 66\%$; employment rate $= 600/1000 = 60\%$. (These are close to recent euro-area orders of magnitude.)

Common pitfall The unemployment rate's denominator is the **labour force**, not the population. So it can fall for a bad reason: when discouraged workers stop searching they leave the labour force entirely and the measured unemployment rate **drops** even though fewer people work. Read it alongside the **employment rate**, whose denominator is fixed, to avoid being fooled.

1.4 What GDP misses

Common pitfall GDP measures **recorded market activity**, not welfare. It omits unpaid household and volunteer work and the informal economy; it ignores the **distribution** of income; it treats environmental damage and resource depletion as free; and it counts some merely-defensive spending (cleaning up a disaster) as gain. Two welfare dimensions it conspicuously fails to capture: **inequality** (GDP can rise while the median household stagnates) and **ecological cost** (a forest felled adds to GDP and subtracts nothing). Necessary information; never the whole story.

2 Inflation and the Phillips curve

Inflation is the sustained rise in the general price level. Its everyday measure is not the GDP deflator but the **consumer price index**, built from the basket a typical household actually buys.

2.1 Constructing the CPI

Worked example — a two-good basket. A household buys 10 loaves and 5 litres of petrol a month. In the base year bread is €2 and petrol €1.50, so the basket costs $10 \times 2 + 5 \times 1.5 =$ €27.50. A year later bread is €2.20 and petrol €1.80: the **same** basket costs $10 \times 2.2 + 5 \times 1.8 =$ €31.00. The CPI is $31.00/27.5 \times 100 = 112.7$, so consumer prices rose 12.7% over the year.

Definition 4 The **consumer price index** (CPI) prices a **fixed basket** of goods, weighted by household spending shares, relative to a base year:

$$\mathrm{CPI}_t = \frac{\text{cost of basket at year-}\ t\ \text{prices}}{\text{cost of basket at base-year prices}} \times 100.$$

CPI **inflation** is its percentage change, $\pi_t = (\mathrm{CPI}_t - \mathrm{CPI}_{t-1})/\mathrm{CPI}_{t-1}$. **Core** inflation strips out the volatile food and energy items to reveal the underlying trend.

Common pitfall Because the basket is **fixed**, the CPI overstates inflation: when petrol jumps, households substitute toward cheaper goods, but the fixed basket keeps buying the expensive one (**substitution bias**). It also struggles with **quality change** and **new goods**. This is why the GDP deflator (which uses current quantities) and the CPI rarely agree exactly — they ask subtly different questions.

The **causes** of inflation are conventionally split into **demand-pull** (aggregate demand outruns the economy's capacity, bidding prices up) and **cost-push** (an input shock — oil in the 1970s, energy in 2022 — raises firms' costs regardless of demand). Its **costs** are real even when anticipated: **menu costs** of changing prices, **shoeleather costs** of economising on cash, distortion of the price signals that coordinate the whole economy, arbitrary **redistribution** from lenders to borrowers, and — when unanticipated — a corrosion of the trust that long contracts depend on.

2.2 The Phillips curve and its mutations

Worked example — the apparent menu. Plotting UK data from the 1950s–60s, A. W. Phillips found a stable downward curve: years of low unemployment were years of high wage inflation, and vice versa. It looked like a **menu** — a government could simply **choose** less unemployment by accepting more inflation. The 1970s then delivered high unemployment **and** high inflation at once (**stagflation**), and the menu vanished. Understanding why is the heart of modern macro policy.

Definition 5 The **original Phillips curve** posits a stable negative relation between inflation π and unemployment u. The **Friedman–Phelps (expectations-augmented)** version adds expected inflation π^e and a natural rate u^*:

$$\pi = \pi^e - \beta(u - u^*).$$

There is a trade-off only for **given** expectations; once workers come to **expect** high inflation, the whole curve shifts up, so any attempt to hold u below u^* buys ever-accelerating inflation, not a permanent jobs gain. In the long run $\pi = \pi^e$, which forces $u = u^*$: the long-run Phillips curve is **vertical**.

Definition 6 The **NAIRU** (non-accelerating-inflation rate of unemployment) is the unemployment rate u^* consistent with **stable** inflation. Below it, inflation accelerates; above it, it decelerates. It is the operational name for the long-run vertical curve's position.

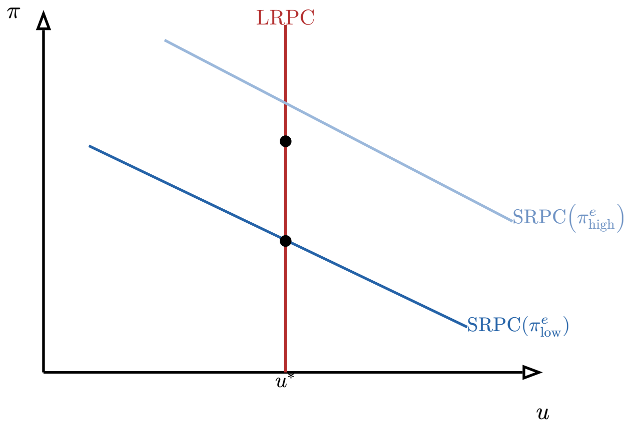

Figure 2: The expectations-augmented Phillips curve. Each downward short-run curve (SRPC) holds expected inflation fixed and offers a temporary trade-off. Pushing unemployment below u^* raises inflation, which raises expectations, shifting the curve up. The long-run curve (LRPC) is vertical at the NAIRU u^*: in the long run there is no trade-off, only the natural rate.

Common pitfall The original curve is not "wrong" so much as **misread as permanent**. It describes a real **short-run** trade-off; the error — which cost the 1970s dearly — is treating it as a stable menu you can sit on forever. Once expectations adjust, the free lunch is gone. This is the empirical limit the OP insists you can state.

3 Economic growth

A recession is a wiggle; **growth** is the trend, and over decades the trend is what determines whether a country is rich or poor. A steady 2% a year doubles living standards in 35 years; 1% takes 70. We ask two questions: how do we **account** for growth that has happened, and what **model** explains it?

3.1 Growth accounting and total factor productivity

Worked example — where did the growth come from?. Output grew 3% last year. Capital grew 2% and hours worked grew 1%. With a capital share of 1/3 and labour share of 2/3, the inputs **alone** explain $1/3 \times 2\% + 2/3 \times 1\% = 0.67\% + 0.67\% = 1.33\%$ of growth. The remaining $3\% - 1.33\% = 1.67\%$ came from neither more machines nor more hours — it is the **residual**, our measure of how much better the economy uses what it has.

Definition 7 With a Cobb–Douglas production function $Y = AK^\alpha L^{1-\alpha}$, growth decomposes (in growth rates, denoted by a hat) as

$$\hat{Y} = \hat{A} + \alpha\hat{K} + (1-\alpha)\hat{L}.$$

The residual $\hat{A} = \hat{Y} - \alpha\hat{K} - (1-\alpha)\hat{L}$ is **total factor productivity** (TFP) growth — the part of output growth not explained by measured inputs. Often called the **Solow residual**, it captures technology, organisation, and efficiency, and is the engine of long-run prosperity.

Common pitfall TFP is a **residual**, so it is also a measure of our **ignorance**: it sweeps up technical progress together with measurement error, mismeasured input quality, and anything else left out. Calling it "technology" is a convenient label, not a direct observation. Treat a TFP number as a question ("what made this happen?") more than an answer.

3.2 The Solow model

Worked example — why saving alone cannot grow you forever. An economy saves a fixed fraction of output and invests it in capital. More capital means more output — but each extra machine adds **less** than the last (diminishing returns, straight from micro), while the capital you already have **wears out** at a constant rate. Eventually the new investment a fixed saving rate can fund is exactly swallowed by depreciation and by equipping a growing workforce. Capital per worker stops rising. The economy reaches a **steady state** — and from there, only better technology can lift output per head.

Definition 8 In the **Solow model**, output per worker $y = f(k)$ depends on capital per worker k, with diminishing returns ($f' > 0$, $f'' < 0$). Capital per worker rises with saving and investment $sf(k)$ and falls with depreciation and labour-force growth $(n+\delta)k$:

$$\Delta k = sf(k) - (n+\delta)k.$$

The **steady state** k^* solves $sf(k^*) = (n+\delta)k^*$: investment just covers depreciation plus equipping new workers, so k — and hence y — is constant. A higher saving rate raises the **level** of k^* but not the long-run **growth rate**, which (for output per worker) is zero without technical progress.

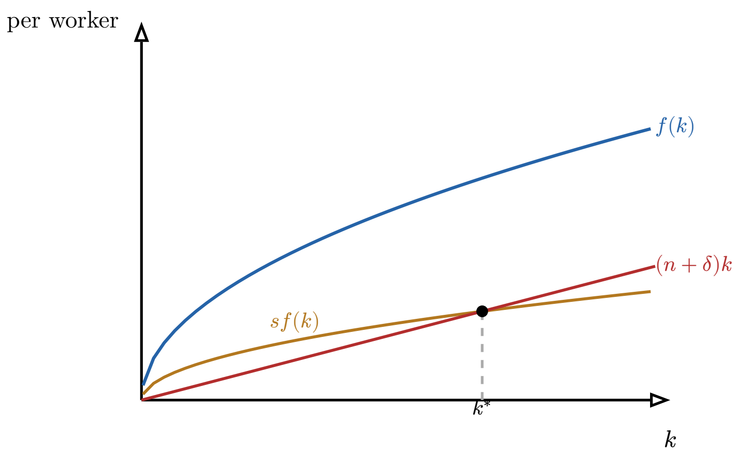

Figure 3: The Solow diagram. Output per worker $f(k)$ is concave. Investment $sf(k)$ (a fixed fraction of it) is compared with the break-even line $(n+\delta)k$ needed to keep capital per worker constant. Where they cross is the steady state k^*: to the left investment exceeds break-even and k rises; to the right it falls. The economy converges to k^* from any starting point.

Definition 9 The **golden-rule** capital stock k_{gold} is the steady state that maximises **consumption** per worker, $c = f(k) - (n+\delta)k$. It occurs where the marginal product of capital equals the break-even rate:

$$f'(k_{\text{gold}}) = n + \delta.$$

Saving **more** than the golden-rule rate is **dynamically inefficient** — it builds capital whose extra output is entirely eaten by the depreciation it creates, leaving less to consume. More saving is not always better.

Definition 10 **Conditional convergence**: because of diminishing returns, a country starting **below** its own steady state grows **faster** (capital is scarce, so its marginal product — and the payoff to investing — is high). Poor countries thus tend to catch up to rich ones **conditional on** sharing the same steady-state parameters (saving, population growth, technology). Where those parameters differ, countries converge to **different** steady states and need not catch up at all.

4 Endogenous growth and sustainability

The Solow model's sharpest lesson is also its great limitation: long-run growth in output per head comes **only** from technical progress A — yet Solow leaves A **exogenous**, an unexplained gift falling from outside. **Endogenous growth theory** asks where A comes from, and sustainability analysis asks what bounds growth must respect.

Definition 11 **Endogenous growth** makes the growth rate emerge from **choices inside the model**. In **Romer-style** theory, ideas are produced by deliberate R&D and — crucially — are **non-rival**: a blueprint, once created, can be used everywhere at once without being used up. Non-rivalry defeats diminishing returns at the level of the whole economy, so deliberate investment in **ideas** can sustain growth indefinitely, in a way investment in **machines** (Solow) cannot.

Definition 12 **Human capital** — the skills and knowledge embodied in workers — enters as a third accumulable factor alongside physical capital and raw labour. Like ideas, it can generate spillovers (an educated workforce makes everyone more productive), which is a further channel through which policy (schooling, training) bears on long-run output.

Remark **Secular stagnation** is the opposing worry: that advanced economies face persistently weak demand and low growth — ageing populations, falling investment demand, a savings glut — so that the natural real interest rate may sit near or below zero, leaving conventional monetary policy chronically short of room. It is the live counter-thesis to techno-optimism.

4.1 Sustainability: the Kaya identity and decoupling

Worked example — four dials on emissions. A country's carbon emissions did not change last year, yet its economy grew 2%. How? Emissions are the product of four factors, and a fall in one offset the rise in another. The bookkeeping that makes this precise is the Kaya identity.

Definition 13 The **Kaya identity** decomposes carbon emissions into four multiplicative drivers:

$$CO_2 = POP \times \frac{GDP}{POP} \times \frac{Energy}{GDP} \times \frac{CO_2}{Energy},$$

i.e. population, output per person (affluence), **energy intensity** of output, and the **carbon intensity** of energy. It turns "cut emissions" into four concrete levers, two of which (the last two) are about **technology and energy mix** rather than about shrinking the economy.

Definition 14 **Relative decoupling** occurs when emissions grow **more slowly** than GDP (intensity falls, but emissions still rise). **Absolute decoupling** occurs when GDP grows while emissions **fall**. Only absolute decoupling is consistent with both growth and a falling carbon stock; whether it can be achieved fast and broadly enough is the central empirical question of growth sustainability.

5 Money and the quantity theory

We now turn from the real economy's long run to the monetary system that overlays it. The first surprise: most money is not printed by the state — it is **created by commercial banks** when they lend.

5.1 Endogenous money and the aggregates

Worked example — a loan creates a deposit. You walk into a bank and are granted a €10 000 loan. The bank does not hand you someone else's savings from a vault; it **credits your account** with €10 000 — a new deposit that did not exist a moment before. The loan (an asset) and the deposit (a liability) appear together on the bank's balance sheet. New money has been created by the act of lending itself. When you repay, that money is destroyed.

Definition 15 **Endogenous money creation**: in a modern economy, the bulk of the money supply is **bank deposits created when banks extend credit**, constrained by capital and regulation and by the central bank's policy rate — not by a fixed stock of reserves multiplied up. (The older **credit multiplier** story — reserves $\times$ 1/reserve-ratio — is a useful first approximation but reverses the true causation: loans drive deposits, which drive reserves, not the other way round.)

Definition 16 Money is measured in nested **aggregates**, from narrow to broad:

- **M1**: currency in circulation + overnight deposits — the most liquid, spendable money.
- **M2**: M1 + deposits redeemable at up to 3 months' notice or with maturity up to 2 years.
- **M3**: M2 + repurchase agreements, money-market fund shares, and short debt securities — the broadest, and the ECB's traditional reference aggregate.

Each layer adds assets that are slightly less liquid than the last.

5.2 The quantity theory of money

Worked example — doubling the money, nothing else. Suppose overnight every euro in the economy magically becomes two, while the number of goods and people's habits are unchanged. There are twice as many euros chasing the same goods bought equally often — so,

> in the long run, every price simply doubles. Output and the real economy are untouched; only the price **level** moved. That intuition is the quantity theory.

Definition 17 The **quantity theory of money** starts from the accounting identity

$$MV = PY,$$

money stock M times its **velocity** of circulation V equals the price level P times real output Y (so PY is nominal GDP). Adding two **assumptions** — that V is roughly stable and that Y is set by real factors (the supply side), independent of M — turns the identity into a **theory**: changes in M pass through one-for-one into P. In growth rates, $\widehat{M} + \hat{V} = \hat{P} + \hat{Y}$, so inflation $\hat{P} \approx \widehat{M} - \hat{Y}$ when velocity is stable. This is the long-run case for "inflation is always and everywhere a monetary phenomenon."

Common pitfall The identity $MV = PY$ is **always** true — it defines V. The **theory** adds the assumptions that V is stable and Y is exogenous, and those **fail** exactly when you most want them: in the short run Y moves with demand, and V is far from constant. After 2008, central banks expanded M enormously while V **collapsed** (the new money sat idle as reserves), so inflation did **not** follow — a textbook refutation of the naïve quantity theory as a short-run guide. It is a long-run proposition, not a forecasting rule.

6 Central banking

A central bank is the institution that manages the monetary system in pursuit of a legal **mandate**. How it is structured, and what tools it wields, determine how monetary policy reaches the real economy — and your firm.

6.1 Mandates and independence

Definition 18 A central bank's **mandate** is its statutory objective. The mandates differ in a way worth knowing: the **ECB** has a **primary** objective of price stability (defined as 2% inflation over the medium term), with other goals strictly subordinate; the **Federal Reserve** has a **dual mandate** — maximum employment **and** stable prices — weighing both; the **Swiss National Bank** (SNB) targets price stability while paying close attention to the franc's exchange rate.

Definition 19 **Central-bank independence** means monetary decisions are insulated from short-term political control. Its rationale is the **time-inconsistency** problem: a government that controls money is always tempted to engineer a pre-election boom, which rational agents anticipate, so that discretion delivers higher inflation with no lasting gain in output. Delegating to an independent, inflation-focused central bank makes the commitment to low inflation **credible**, which is itself what anchors expectations — closing the loop back to the Phillips curve.

6.2 Conventional and unconventional tools

Definition 20 The **conventional** instrument is the short-term **policy interest rate** (the ECB's deposit facility rate, the Fed funds target, the SNB policy rate). Raising it makes borrowing dearer and saving more attractive, cooling demand and inflation; cutting it does the reverse. Supporting tools are **reserve requirements** and **open-market operations** (buying or selling short-term securities to steer the rate toward target).

Definition 21 When the policy rate hits its **effective lower bound** (near zero) and more easing is needed, central banks turn to **unconventional** tools:

- **Quantitative easing (QE)**: large-scale purchases of longer-term bonds, creating reserves to push **long** rates down and ease financial conditions once the **short** rate can fall no further.
- **Forward guidance**: communicating the likely future path of rates to shape expectations today (since long rates are an average of expected future short rates).

Worked example — the 2022–2024 tightening, compared. Facing the post-pandemic energy-and-supply inflation shock, three central banks moved in the same direction at different speeds. The **Fed** lifted its funds target from near zero in early 2022 to a range of about 5.25--5.5% by mid-2023. The **ECB** raised its deposit rate from −0.5% in mid-2022 to 4.0% by September 2023 — exiting eight years of negative rates. The **SNB**, with lower domestic inflation (helped by a strong franc that cheapens imports), peaked at only 1.75% and was the **first** major central bank to **cut**, in March 2024. Same shock, three mandates and three starting points, three distinct paths — and three different signals for any firm financing itself in euros, dollars, or francs.

6.3 Transmission to business

Definition 22 The **monetary transmission mechanism** is the chain from a policy-rate change to the real economy. A rate rise propagates through several channels:

- **Interest-rate / investment channel**: higher rates raise the hurdle a project's return must clear, so firms' investment I falls.
- **Credit channel**: dearer and scarcer bank lending squeezes borrowers.
- **Exchange-rate channel**: higher domestic rates attract capital, **appreciating** the currency, which cheapens imports (cooling inflation) but hurts exporters.
- **Asset-price / wealth channel**: higher rates depress bond and equity prices, reducing wealth and spending.

Common pitfall Make the **business translation** explicit, because the later finance courses demand it. A central-bank rate hike raises your firm's **cost of debt**, and through it the **weighted average cost of capital** (WACC) you will use to discount projects in **BUS33-3** — so fewer projects clear the bar. It also raises the financing cost of **working capital** (the BFR of **BUS23-3**): every euro tied up in inventory and receivables now costs more to fund. A rate decision is never abstract; it is a line item in your next budget.

7 Fiscal policy

The other great lever is the government's own budget — its spending G and taxes T. Its short-run power rests on a single mechanism that amplifies an initial impulse: the **multiplier**.

7.1 The Keynesian multiplier

> **Worked example — the ripple from €1 of spending.** The government spends an extra €100 on a bridge. The builders earn €100 and, if they spend 80% of any extra income, spend €80 — which becomes someone else's income, who spends €64, and so on: $100 + 80 + 64 + 48.8 + \ldots$. The sum is not €100 but $100/(1 - 0.8) =$ €500. One euro of spending raised total demand by **five**. That amplification is the multiplier.

Definition 23 Let the **marginal propensity to consume** be c (the fraction of an extra euro of income that is spent), so consumption is $C = C_0 + c(Y - T)$. In the simplest closed economy, equilibrium output is $Y = C + I + G$, and solving gives the **government-spending multiplier**

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1-c}.$$

With $c = 0.8$ the multiplier is 5; with $c = 0.6$ it is 2.5. The **tax multiplier** is smaller in magnitude and negative, $-c/(1-c)$, because a tax cut first has to be partly **saved** before any of it is spent.

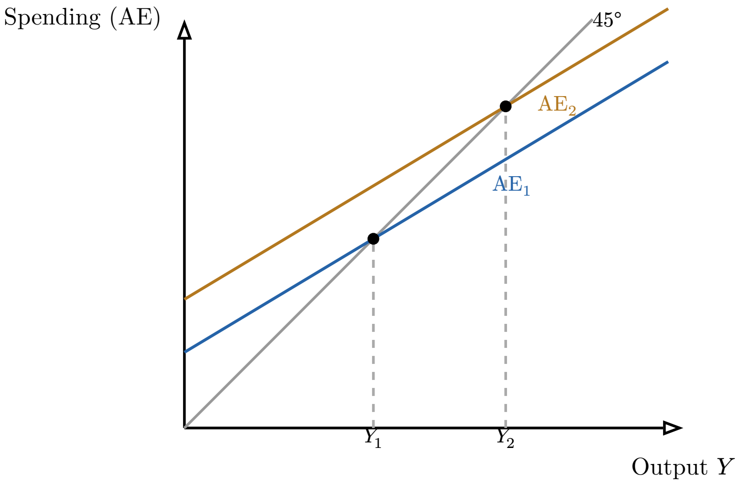

Figure 4: The Keynesian cross. Equilibrium output is where planned spending (AE) crosses the 45° line (AE $= Y$). An upward shift of AE by ΔG raises equilibrium output by the **larger** amount $\Delta Y = \Delta G/(1-c)$ — the horizontal gap $Y_2 - Y_1$ exceeds the vertical shift. That ratio is the multiplier.

7.2 Automatic stabilisers, discretion, and crowding out

Definition 24 **Automatic stabilisers** are features of the budget that cushion the cycle **without any new decision**: in a downturn, tax revenue falls and unemployment benefits rise automatically, supporting demand; in a boom the reverse restrains it. **Discretionary** policy is a deliberate, legislated change in G or T. Stabilisers are timely but limited; discretion is powerful but subject to **recognition, decision, and implementation lags** that can make it arrive too late.

Definition 25 The **crowding-out effect** is the offset that limits fiscal stimulus: government borrowing to fund a deficit raises the demand for loanable funds, pushing **interest rates up**, which **reduces** private investment I. Part of the demand the government adds is thus subtracted from the private sector — so the realised multiplier is smaller than the simple $1/(1-c)$. The full account of **how much** smaller needs both the goods market and the money market together — which is exactly IS–LM.

8 The IS–LM model

We have a goods market (where fiscal policy and the multiplier live) and a money market (where the central bank and the interest rate live). IS–LM is the keystone that holds them in **simultaneous** equilibrium, on one diagram of the interest rate r against output Y. It is the single most useful short-run picture in the course.

8.1 The two curves

Definition 26 The **IS curve** is the set of (Y, r) pairs at which the **goods market** clears (investment = saving). It slopes **downward**: a lower interest rate raises investment I, which — through the multiplier — raises equilibrium output Y. Its **position** depends on fiscal policy: higher G or lower T shifts IS **rightward** (by ΔG times the multiplier).

Definition 27 The **LM curve** is the set of (Y, r) pairs at which the **money market** clears (money demand = money supply). It slopes **upward**: higher output raises the demand for money (more transactions), and with a fixed money supply that pushes the interest rate **up**. Its **position** depends on monetary policy: an increase in the money supply shifts LM **rightward (down)**, lowering r at each Y.

Worked example — solving a small IS–LM. Suppose $\text{IS} : Y = 1000 - 20r$ and $\text{LM} : Y = 400 + 30r$ (with r in percentage points). Equilibrium sets them equal: $1000 - 20r = 400 + 30r \Rightarrow 600 = 50r \Rightarrow r^* = 12$, and $Y^* = 1000 - 20 \times 12 = 760$. A fiscal stimulus that shifts IS to $Y = 1120 - 20r$ gives $r^* = 14.4$, $Y^* = 832$: output rises by 72, **not** by the full 120 of the IS shift — the rest is crowded out by the higher interest rate. IS–LM **quantifies** crowding out.

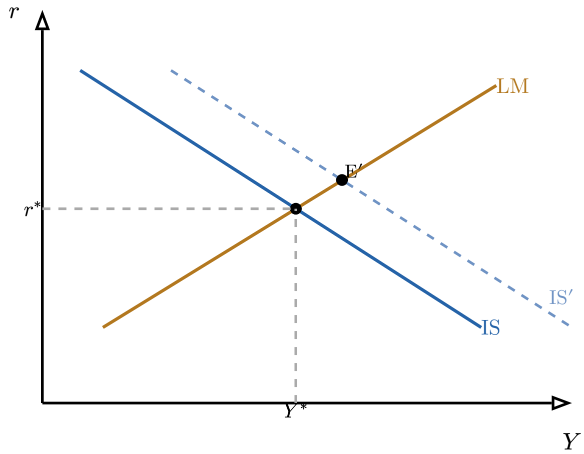

Figure 5: IS–LM. The economy sits where the downward IS (goods market) meets the upward LM (money market). A fiscal expansion shifts IS right to IS′: output and the interest rate both rise. Because the higher r chokes off some investment, output rises by **less** than the IS shift — the visible measure of crowding out.

8.2 Recessionary and inflationary gaps

Definition 28 Compare equilibrium output Y^* with **potential output** Y_p (the full-employment level, where the economy produces at its sustainable capacity). A **recessionary gap** exists when $Y^* < Y_p$: idle resources and unemployment above the NAIRU, calling for **expansionary** policy (IS right via fiscal stimulus, or LM right via monetary easing). An **inflationary gap** exists when $Y^* > Y_p$: the economy overheats and inflation accelerates, calling for **contractionary** policy.

Common pitfall IS–LM is a **short-run, fixed-price** model — its power and its limit. It explains how demand-side policy moves output and interest rates when prices are sticky, but it says nothing on its own about the long-run vertical Phillips curve or supply shocks. Use it to read a recession or a stimulus; do not ask it to deliver long-run growth, which belongs to Solow, not IS–LM. Knowing which model answers which question is the skill.

9 Debt and GDP dynamics

Fiscal stimulus is financed by borrowing, and borrowing accumulates into **public debt**. Whether a debt is sustainable is not about its raw size but about its **trajectory relative to GDP** — a dynamic governed by one equation.

> **Worked example — can you grow out of debt?.** A country owes 100% of GDP, pays a 3% interest rate on it, and runs a balanced **primary** budget (revenue equals spending before interest). If the economy grows at 3%, the debt ratio holds steady — growth exactly matches the interest piling up. If growth slows to 1% while the rate stays at 3%, the ratio **climbs** even with a balanced primary budget: the debt outgrows the economy. The gap between the interest rate and the growth rate, $r - g$, is the whole story.

Definition 29 Let d be the debt-to-GDP ratio, r the (real) interest rate on debt, g the (real) GDP growth rate, and p the **primary balance** (surplus, as a share of GDP, before interest). The debt ratio evolves as

$$\Delta d \approx (r - g)d - p.$$

The sign of $r - g$ is decisive. If $r > g$, debt has a built-in tendency to **snowball** and a primary **surplus** $p = (r - g)d$ is needed merely to **stabilise** it. If $r < g$, the ratio shrinks on its own even with modest primary deficits. There is no single universal "critical threshold" — proposed lines such as 90% are empirically fragile — but a sustained $r > g$ with persistent primary deficits is the configuration that makes any debt level dangerous.

Common pitfall Do not read a fixed debt-ratio number (60%, 90%, 100%) as a cliff edge. Japan sustains a ratio above 200% at low rates; other countries face crises far below 100%. Sustainability is about **dynamics** — the $r - g$ gap and the primary balance — and about **who holds the debt and in what currency**, not about crossing a magic line. The famous "90% threshold" result did not survive scrutiny; treat thresholds as rules of thumb, not laws.

10 Policy-mix case studies

We close by running the whole apparatus — IS–LM, the multiplier, monetary tools, debt dynamics — on the three defining macro shocks of the last two decades. The point is not the history but the **diagnosis**: which curve moved, which lever answered, what worked, and why.

Definition 30 A **policy mix** is the combination of fiscal and monetary policy deployed against a shock. Reading one means asking, in order: (1) was the shock to **demand** or **supply**? (2) which way did IS and LM move? (3) what did fiscal and monetary policy do, and were they pulling together or against each other? (4) what did the debt dynamics and inflation outcome reveal about limits?

Worked example — 2008 — the Global Financial Crisis (a demand collapse). A financial panic crashed credit and confidence: a sharp **leftward** shift of IS (collapsing I and C), driving a deep recessionary gap. Monetary policy cut rates to the zero lower bound and, with conventional room exhausted, turned to **QE** — the first large use of the unconventional toolkit. Fiscal stimulus (e.g. the US ARRA) added demand. **What worked**: catastrophe was averted and the system stabilised. **What failed**: the recovery was slow, partly because fiscal support was withdrawn too early (a premature pivot to **austerity** in parts of Europe), and the quantity-theory prediction of inflation from QE **did not materialise** — velocity collapsed, the new reserves sat idle.

Worked example — 2020 — COVID-19 (a joint supply-and-demand shock). Lockdowns hit **both** sides at once: supply (closed factories, broken chains) and demand (collapsed spending). The response was the largest **coordinated** policy mix in peacetime — massive fiscal transfers (furlough schemes, direct payments) shifting IS hard right, alongside renewed QE and near-zero rates holding LM down. **What worked**: incomes and firms were bridged through the shutdown, avoiding a 1930s-style spiral; debt ratios jumped but rates stayed low ($r < g$), keeping the dynamics manageable. **What failed / the cost**: the combination of supply constraints, pent-up demand released at reopening, and the energy shock to come set the stage for the inflation of 2022.

Worked example — 2022 — the inflation shock (a supply-side surge in prices). Post-reopening demand met constrained supply and an energy-price spike (war in Ukraine): a **cost-push** inflation surge, the worst in forty years. Here fiscal and monetary policy were pulled in **different** directions — central banks tightened hard (the Fed/ECB/SNB sequence above) to bring inflation down, while governments often **cushioned** households with energy subsidies, partly offsetting the monetary brake. **What worked**: aggressive, credible tightening re-anchored expectations and brought inflation down without (so far) a deep recession — a "soft landing." **What it revealed**: the limits of demand-management tools against a **supply** shock (you cool inflation only by also cooling output), and the fiscal– monetary tension a policy mix can contain.

The thread of this book. Macroeconomics is micro's marginal reasoning rebuilt at the level of the whole system. First **measure** — GDP three ways, real versus nominal, the labour-market rates, CPI — and remember every measure hides something. Then the **long run** is about growth: accounting splits it into inputs and the TFP residual; Solow shows capital alone hits a steady state, so only ideas and human capital (endogenous growth) sustain it, within sustainability bounds the Kaya identity makes concrete. The **short run** is about demand and the two levers: **money**, created endogenously by banks and governed by a central bank whose rate decisions transmit straight into your WACC and BFR; and **fiscal** policy, amplified by the multiplier but limited by crowding out. **IS–LM** unites them, locating output and the interest rate and revealing recessionary and inflationary gaps; **debt dynamics** $(r - g)$ bound the fiscal lever. Put together on the 2008, 2020, and 2022 episodes, the whole framework lets you read any central-bank or budget decision as a concrete business signal — which is what every finance and strategy course after this one will assume you can do.

Exercises

1. Take a small open economy with given output sold to households, firms, and abroad. Compute GDP by all three approaches (production/value added, expenditure $C + I + G + (X - M)$, income) and verify they agree. Then, given current- and base-year prices, compute nominal GDP, real GDP, and the GDP deflator, and identify at least two welfare dimensions GDP fails to capture.

2. From a table of population, employment, unemployment, and inactivity, compute the unemployment, activity, and employment rates. Then show numerically how the unemployment rate can **fall** when discouraged workers leave the labour force, and explain why the employment rate is the safer gauge.

3. Given a fixed consumption basket and two years of prices, construct the CPI and compute the inflation rate. Then write the expectations-augmented Phillips curve, explain the difference between the original, Friedman-augmented, and NAIRU formulations, and use the vertical long-run curve to explain why a permanent inflation–unemployment trade-off does not exist.

4. Using growth accounting with a Cobb–Douglas function, decompose a given output growth rate into contributions from capital, labour, and total factor productivity, and interpret the TFP residual — including why it is a measure of ignorance as much as of technology.

5. For a Solow economy with a stated production function, saving rate, depreciation, and labour-force growth, solve for the steady-state capital and output per worker, find the golden-rule saving rate, and explain conditional convergence — why a poorer economy with the same parameters grows faster.
6. Contrast exogenous (Solow) and endogenous (Romer-style) growth: explain how non-rival ideas and human capital can sustain long-run growth that capital accumulation alone cannot. Then use the Kaya identity to decompose a change in a country's emissions and distinguish relative from absolute decoupling.
7. Explain endogenous money creation with a bank balance sheet, define M1/M2/M3, and evaluate the quantity theory $MV = PY$: state its assumptions, derive the long-run inflation implication, and give one concrete episode (e.g. post-2008 QE) in which it failed and why.
8. Compare the ECB, Fed, and SNB mandates and their 2022–2024 rate paths. Then trace the transmission of a 1-point rate hike through the investment, credit, and exchange-rate channels, and translate it into its effect on a specific firm's WACC and BFR financing cost.
9. Derive the government-spending and tax multipliers from a consumption function with marginal propensity to consume c, apply them numerically to a stated fiscal stimulus, and explain how automatic stabilisers and crowding out modify the realised effect.
10. Given algebraic IS and LM curves, solve for equilibrium output and the interest rate; impose a fiscal expansion and a monetary expansion in turn, computing the new equilibrium each time and measuring the crowding-out; and identify whether the economy faces a recessionary or an inflationary gap relative to a stated potential output.
11. Using the debt-dynamics equation $\Delta d \approx (r - g)d - p$, compute the primary balance required to stabilise a given debt ratio under stated r and g, analyse a real-data scenario in which $r - g$ flips sign, and explain why fixed debt thresholds are unreliable.
12. Pick one of 2008, 2020, or 2022. Using IS–LM plus debt dynamics, identify the shock (demand or supply), the direction each curve moved, the fiscal and monetary response, and evaluate what worked, what failed, and why.